

Muhammad Zain

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PROFILE

Final-year Software Engineering student seeking an 8 to 12 month co-op starting September 2026. Shipped a full-stack web app to production scoring 99 on Lighthouse, and built machine learning pipelines on real multi-year climate data.

EDUCATION

University of Calgary

Bachelor of Science (Software Engineering)

Calgary, AB

September 2022 - April 2027

SKILLS

- **Programming Languages:** Python, Java, C++, JavaScript, TypeScript, SQL, HTML, CSS
- **Backend & APIs:** Flask, REST APIs, PostgreSQL, MySQL, JWT + OAuth, JSON, LLM Integration, Gemini API
- **Frontend & UI:** React, Three.js, WebGL, Vite, Responsive Design
- **DevOps & Tooling:** Git, GitHub Actions (CI/CD), Docker, Docker Compose, GCP (Cloud Run, Cloud SQL, GCS), Jira, Agile

RELEVANT EXPERIENCE

Enactus Canada

Software Development Intern

Calgary, AB

September 2025 - May 2026

- Designed a spatiotemporal data pipeline in sprints, aligning ERA5 climate reanalysis, satellite vegetation indices, storm events, & static terrain features onto a canonical 30 km grid to produce weekly wildfire risk scores across Alberta.
- Enforced leakage-safe temporal validation & nearest-neighbor grid joins through peer code review, eliminating coordinate drift, record duplication, & forward-looking time contamination across multi-year fire-season experiments.
- Surfaced 40-60% of fires within the top 1% of ranked cells under 500:1 extreme class imbalance, tuning gradient-boosted models for PR-AUC & Recall@Top-K & adding an Isolation Forest anomaly score agreed in design review.
- Standardized reproducible outputs including Parquet tables, serialized model artifacts, & interactive risk-map visualizations, presenting findings to technical & non-technical stakeholders for operational handoff.

PROJECTS

Resume & Career Coach (RCC)

January 2026 - June 2026

Python, Flask, React, PostgreSQL, Docker, Gemini LLM

- Shipped [resumecoach.app](#), a live production app scoring resume-to-posting fit via a hybrid token-overlap + LLM pipeline.
- Scheduled 189 consecutive keep-warm CI runs at a 0% failure rate, mitigating cold starts on a free-tier host.
- Engineered anti-hallucination grounding via structured JSON prompting, curated skill coverage, & SHA-256 caching.
- Secured 22 REST endpoints with JWT access & refresh tokens, Google OAuth, email verification, & rate limiting.

ProfileHub Performance Evaluation

January 2026 - April 2026

Python, Flask, React, MySQL, GCP, Locust, SciPy

- Reached 0% request failures across 72 load runs by fixing a connection-pool leak that drained slots under load.
- Measured 2.4x higher throughput on read-heavy versus write-heavy workloads, 5.12 against 2.14 requests per second.
- Identified non-linear saturation past 15 concurrent users, a 38% latency jump that set the team's capacity ceiling.
- Deployed a containerized service on Cloud Run with managed SQL, signed-URL object storage, & a 4-worker runtime.

Lumen Pendulum

October 2025 - November 2025

TypeScript, Three.js, WebGL, Vite, GitHub Actions

- Delivered a public interactive 3D visualization holding 60fps under continuous animation & direct drag control.
- Removed frame stutter from garbage collection with a zero-allocation render loop over pre-allocated GPU buffers.
- Implemented real-time physics from Lagrangian mechanics with a 4th-order Runge-Kutta integrator at 8 sub-steps.

CityX Subway Display System

June 2025 - July 2025

Java, Swing, MySQL, Multithreading, REST APIs

- Sustained 90% display reliability under flaky upstream APIs with 15s timeouts, 3-attempt retry/backoff, & caching.
- Kept a 120-station transit network updating in real time via multithreaded pipelines in clean MVC architecture.
- Consolidated 3 external data feeds into one normalized model the team built against as a single source of truth.